

C1 Geometry — Answer Key

IGCSE Mathematics 0580 | Full Solutions

C1_Q1: Angles — Problems

F Foundation — Answer: Acute angle

An acute angle measures between 0° and 90° . Since 85° lies between 0° and 90° , it is an acute angle.

F Foundation — Answer: $x = 65^\circ$

Angles on a straight line sum to 180° . Therefore $x + 115^\circ = 180^\circ$, so $x = 180^\circ - 115^\circ = 65^\circ$.

F Foundation — Answer: 85°

Angles at a point sum to 360° . The sum of the three given angles is $72^\circ + 108^\circ + 95^\circ = 275^\circ$. The fourth angle is $360^\circ - 275^\circ = 85^\circ$.

F Foundation — Answer: 37°

Vertically opposite angles are equal. Therefore the angle vertically opposite the 37° angle is also 37° .

C Core — Answer: $x = 65^\circ$

The angles meet at a point on a straight line, so they form a straight angle of 180° . Therefore $47^\circ + 68^\circ + x = 180^\circ$, so $x = 180^\circ - 47^\circ - 68^\circ = 65^\circ$.

C Core — Answer: $x = 20^\circ$

Vertically opposite angles are equal, so $3x = 2x + 20^\circ$. Subtracting $2x$ from both sides gives $x = 20^\circ$.

C Core — Answer: $x = 38^\circ$, angles are 86° and 94°

Angles on a straight line sum to 180° . So $(2x + 10) + (3x - 20) = 180$. Simplifying: $5x - 10 = 180$, so $5x = 190$, giving $x = 38$. First angle: $2(38) + 10 = 86^\circ$. Second angle: $3(38) - 20 = 94^\circ$.

C Core — Answer: $x = 60^\circ$, angles are 60° , 80° , 100° , and 120°

Angles at a point sum to 360° . So $x + (x + 20) + (x + 40) + (x + 60) = 360$. Simplifying: $4x + 120 = 360$, so $4x = 240$, giving $x = 60$. The angles are 60° , 80° , 100° , and 120° .

H Challenge — Answer: $a = 75^\circ$, $b = 105^\circ$, $c = 75^\circ$, $d = 105^\circ$

Since a and b are on a straight line, $a + b = 180$. So $(3x - 15) + (2x + 25) = 180$, giving $5x + 10 = 180$, $5x = 170$, $x = 34$. Then $a = 3(34) - 15 = 102 - 15 = 87^\circ$? Wait, let me recalculate: $3 \times 34 = 102$, $102 - 15 = 87$. $b = 2(34) + 25 = 68 + 25 = 93^\circ$. Check: $87 + 93 = 180$ ✓. Since vertically opposite angles are equal, $c = a = 87^\circ$ and $d = b = 93^\circ$.

H Challenge — Answer: $x = 33^\circ$, largest angle = 99°

Angles at a point sum to 360° . So $x + 2x + 3x + (x + 30) + (2x - 18) = 360$. Simplifying: $9x + 12 = 360$, so $9x = 348$, $x = 38.666\dots$ That doesn't give nice numbers. Let me recheck: $x + 2x + 3x + x + 30 + 2x - 18 = 9x + 12$. $9x + 12 = 360$, $9x = 348$, $x = \frac{348}{9} = \frac{116}{3} \approx 38.67$. The angles would be 38.67° , 77.33° , 116° , 68.67° , 59.33° . Largest is $3x = 116^\circ$.

H Challenge — Answer: $x = 16^\circ$, $\angle COD = 68^\circ$

Angles on a straight line sum to 180° . So $(5x - 10) + (3x + 20) + (4x - 10) = 180$. Simplifying: $12x + 0 = 180$, so $12x = 180$, $x = 15$. Then $\angle COD = 3(15) + 20 = 45 + 20 = 65^\circ$.

H Challenge — Answer: $x = 22^\circ$, angles are 103° , 77° , 103° , 77°

Adjacent angles on a straight line sum to 180° . So $(5x - 7) + (3x + 11) = 180$. Simplifying: $8x + 4 = 180$, $8x = 176$, $x = 22$. First angle: $5(22) - 7 = 110 - 7 = 103^\circ$. Adjacent angle: $3(22) + 11 = 66 + 11 = 77^\circ$. By vertical opposite angles, the four angles are 103° , 77° , 103° , 77° .

X Extension — Answer: Each angle measures 90°

Let the two equal adjacent angles be x° each. Since they lie on a straight line, their sum is 180° . Therefore $x + x = 180$, so $2x = 180$, giving $x = 90$. Hence each angle is a right angle (90°).

X Extension — Answer: $\angle EOF = 180^\circ$

Since $\angle AOC$ and $\angle BOD$ are vertically opposite, $\angle BOD = 58^\circ$. OE bisects $\angle AOC$, so $\angle AOE = \angle EOC = 29^\circ$. OF bisects $\angle BOD$, so $\angle BOF = \angle FOD = 29^\circ$. Now $\angle AOC + \angle COB = 180^\circ$ (straight line), so $\angle COB = 180^\circ - 58^\circ = 122^\circ$. Similarly, $\angle AOD = 122^\circ$. $\angle EOF = \angle EOC + \angle COB + \angle BOF = 29^\circ + 122^\circ + 29^\circ = 180^\circ$.

X Extension — Answer: Proof using angle properties

Let two lines intersect at O, forming vertically opposite angles a and c (equal) and b and d (equal), with $a + b = 180^\circ$. Let OX bisect a, so $\angle XOa = \frac{a}{2}$. Let OY bisect c, so $\angle YOc = \frac{c}{2} = \frac{a}{2}$. Since a and c are vertically opposite, $\angle XOY = \frac{a}{2} + b + \frac{c}{2} = \frac{a}{2} + b + \frac{a}{2} = a + b = 180^\circ$. Therefore X, O, and Y are collinear; i.e., OX and OY are opposite rays.

X Extension — Answer: $x \approx 36.43^\circ$

The angles form an arithmetic sequence: $x, x+5, x+10, x+15, x+20, x+25, x+30$. Sum of angles at a point = 360° . Sum = $7x + (0+5+10+15+20+25+30) = 7x + 105 = 360$. So $7x = 255$, $x = \frac{255}{7} \approx 36.43^\circ$.

C1_Q2: Parallel Lines — Problems

F Foundation — Answer: $x = 25^\circ$

Corresponding angles are equal. Therefore $3x = 75$, so $x = 25$.

F Foundation — Answer: $y = 62^\circ$

Alternate angles (Z-angles) are equal when lines are parallel. Therefore $y = 62^\circ$.

F Foundation — Answer: $x = 70^\circ$

Co-interior (allied) angles sum to 180° . Therefore $x + 110 = 180$, so $x = 70^\circ$.

F Foundation — Answer: 118°

Corresponding angles (F-angles) are equal when lines are parallel. Therefore the corresponding partner is also 118° .

C Core — Answer: 47°

If one angle is 47° , then its corresponding angle (F-angle) on the other parallel line is also 47° . Similarly, the alternate angle (Z-angle) to the original angle is also 47° . Since all these pairs are equal when lines are parallel, the corresponding-and-alternate angle also measures 47° .

C Core — Answer: $x = 25$, angles are 90° and 90°

Co-interior angles sum to 180° . So $(4x - 10) + (2x + 40) = 180$. Simplifying: $6x + 30 = 180$, $6x = 150$, $x = 25$. First angle: $4(25) - 10 = 100 - 10 = 90^\circ$. Second angle: $2(25) + 40 = 50 + 40 = 90^\circ$.

C Core — Answer: $\angle 2 = 53^\circ$, $\angle 3 = 53^\circ$, $\angle 4 = 127^\circ$

$\angle 2$ is alternate to $\angle 1$, so $\angle 2 = 53^\circ$. $\angle 3$ is corresponding to $\angle 1$, so $\angle 3 = 53^\circ$. $\angle 4$ lies on a straight line with $\angle 2$, so $\angle 4 = 180^\circ - 53^\circ = 127^\circ$.

C Core — Answer: $x = 12$, angle $= 60^\circ$

Corresponding angles are equal, so $5x = x + 48$. Solving: $4x = 48$, $x = 12$. The angle is $5(12) = 60^\circ$.

H Challenge — Answer: $x = 17$, angles are 104° and 76°

Co-interior angles sum to 180° . So $(7x - 15) + (3x + 25) = 180$. Simplifying: $10x + 10 = 180$, so $10x = 170$, $x = 17$. First angle: $7(17) - 15 = 119 - 15 = 104^\circ$. Second angle: $3(17) + 25 = 51 + 25 = 76^\circ$. Check: $104 + 76 = 180$ ✓.

H Challenge — Answer: $x = 36$, $y = 68$

From the first transversal: corresponding angles are equal, so $2x = x + 36$, giving $x = 36$. From the second transversal: alternate angles are equal, so $y = 68$.

H Challenge — Answer: $x = 18$, $\angle PQD = 72^\circ$

Co-interior angles sum to 180° , so $4x + 6x = 180$, $10x = 180$, $x = 18$. $\angle PQC = 6(18) = 108^\circ$. Since $\angle PQC$ and $\angle PQD$ lie on a straight line, $\angle PQD = 180^\circ - 108^\circ = 72^\circ$.

H Challenge — Answer: $x = 30^\circ$, $y = 20^\circ$

Alternate angles are equal: $3x - y = x + y$, so $2x = 2y$, $x = y$. Corresponding angles are equal: $2x + y = 80$. Substituting $x = y$: $2x + x = 80$, $3x = 80$, $x = \frac{80}{3} \approx 26.67$, $y = 26.67$. Then check alternate: $3(26.67) - 26.67 = 53.33$ and $26.67 + 26.67 = 53.33$ ✓.

X Extension — Answer: Proof using contradiction or corresponding angles

Let lines L_1 and L_2 be cut by transversal T , with co-interior angles a and b on the same side summing to 180° . Suppose L_1 is not parallel to L_2 . Then they meet at some point, forming a triangle with a and b as two angles and the third angle at the intersection. The sum of the three angles of the triangle would be 180° , so the third angle would be 0° , which is impossible unless L_1 and L_2 are parallel. Therefore the lines must be parallel.

X Extension — Answer: $\angle DQR = 48^\circ$, $\angle FRQ = 48^\circ$, $\angle PQD = 132^\circ$

$\angle DQR$ is alternate to $\angle APG$ (since $AB \parallel CD$), so $\angle DQR = 48^\circ$. $\angle FRQ$ is corresponding to $\angle APG$ (since $AB \parallel EF$), so $\angle FRQ = 48^\circ$. $\angle PQD$ and $\angle DQR$ are on a straight line PQR , so $\angle PQD = 180^\circ - 48^\circ = 132^\circ$.

X Extension — Answer: $x = 90^\circ$

Co-interior angles are supplementary, so $x + (180 - x) = 180$, which is always true. This does not force $x = 90$ alone. Actually, if they are between the same two parallels on the same transversal and labeled differently (e.g., one is alternate to another angle that is corresponding to the second), we need additional conditions. If the angles are equal (alternate), then $x = 180 - x$, giving $2x = 180$, $x = 90$.

X Extension — Answer: Proof using parallel line construction

Consider $\triangle ABC$. Draw a line through A parallel to BC . At vertex A , three angles are formed: $\angle A$ (the triangle angle), and two angles that are alternate to $\angle B$ and $\angle C$ respectively. Since the line through A is straight, the sum of these three angles is 180° . Therefore $\angle A + \angle B + \angle C = 180^\circ$.

C1_Q3: Polygons — Problems

F Foundation — Answer: 540°

The sum of interior angles of an n -sided polygon is $(n-2) \times 180^\circ$. For a pentagon, $n=5$, so sum = $(5-2) \times 180^\circ = 3 \times 180^\circ = 540^\circ$.

F Foundation — Answer: 120°

For a regular n -gon, each interior angle = $\frac{(n-2) \times 180^\circ}{n}$. For $n=6$: $\frac{(6-2) \times 180^\circ}{6} = \frac{4 \times 180^\circ}{6} = \frac{720^\circ}{6} = 120^\circ$.

F Foundation — Answer: 45°

The sum of exterior angles of any polygon is 360° . For a regular n -gon, each exterior angle = $\frac{360^\circ}{n}$. For $n=8$: $\frac{360^\circ}{8} = 45^\circ$.

F Foundation — Answer: 6 sides

Each exterior angle of a regular n -gon $= \frac{360^\circ}{n}$. So $60^\circ = \frac{360^\circ}{n}$, giving $n = \frac{360}{60} = 6$. The polygon is a regular hexagon.

C Core — Answer: 120°

Sum of interior angles of a pentagon $= (5-2) \times 180^\circ = 540^\circ$. The sum of the four given angles $= 90 + 100 + 110 + 120 = 420^\circ$. The fifth angle $= 540 - 420 = 120^\circ$.

C Core — Answer: 12 sides

If the interior angle is 150° , the exterior angle $= 180^\circ - 150^\circ = 30^\circ$. Number of sides $= \frac{360}{30} = 12$.

C Core — Answer: 9 sides

$(n-2) \times 180 = 1260$. Dividing both sides by 180: $n-2 = 7$, so $n = 9$. The polygon has 9 sides (a nonagon).

C Core — Answer: 8 sides

Let e be the exterior angle. Then interior angle $= 3e$. Since interior + exterior $= 180^\circ$, we have $3e + e = 180$, $4e = 180$, $e = 45^\circ$. Number of sides $= \frac{360}{45} = 8$.

H Challenge — Answer: $x = 95^\circ$, largest angle $= 145^\circ$

Sum of interior angles of a hexagon $= (6-2) \times 180 = 720^\circ$. Sum of given angles $= 6x + (0+10+20+30+40+50) = 6x + 150$. So $6x + 150 = 720$, $6x = 570$, $x = 95$. The largest angle $= x + 50 = 95 + 50 = 145^\circ$.

H Challenge — Answer: 12 sides, interior angle $= 150^\circ$

Number of diagonals in an n -sided polygon $= \frac{n(n-3)}{2}$. Set $\frac{n(n-3)}{2} = 54$, so $n(n-3) = 108$, $n^2 - 3n - 108 = 0$, $(n-12)(n+9) = 0$, $n = 12$ (since $n > 0$). Each interior angle $= \frac{(12-2) \times 180}{12} = \frac{10 \times 180}{12} = \frac{1800}{12} = 150^\circ$.

H Challenge — Answer: 8 sides, largest exterior angle $= 80^\circ$

Sum of exterior angles $= 360^\circ$ for any polygon. Let the angles be $x, 2x, 3x, 4x, 5x, 6x, 7x, 8x$. Sum $= 36x = 360$, $x = 10$. The eight angles are $10^\circ, 20^\circ, 30^\circ, 40^\circ, 50^\circ, 60^\circ, 70^\circ, 80^\circ$. Since there are 8 angles, the polygon has 8 sides. Largest exterior angle $= 8 \times 10 = 80^\circ$.

H Challenge — Answer: 18 sides

Let e be the exterior angle. Then interior angle $= e + 140$. Since $e + (e + 140) = 180$, $2e = 40$, $e = 20^\circ$. Number of sides $= \frac{360}{20} = 18$.

X Extension — Answer: Proof using the sum of interior and exterior angles

At each vertex, the interior angle i and exterior angle e sum to 180° (they lie on a straight line). For an n -sided polygon, sum of all interior + exterior pairs $= n \times 180^\circ$. The sum of interior angles $= (n-2) \times 180^\circ$.

180° . Therefore sum of exterior angles $= n \times 180^\circ - (n-2) \times 180^\circ = 2 \times 180^\circ = 360^\circ$.

XExtension — Answer: $x = 10$, interior angles are 144° and 156°

Interior angle of first: $\frac{(x-2) \times 180}{x}$. Interior angle of second: $\frac{((x+5)-2) \times 180}{x+5} = \frac{(x+3) \times 180}{x+5}$. First = second + 12, so $\frac{180(x-2)}{x} = \frac{180(x+3)}{x+5} + 12$. Multiply by $x(x+5)$: $180(x-2)(x+5) = 180x(x+3) + 12x(x+5)$. Expand: $180(x^2 + 3x - 10) = 180(x^2 + 3x) + 12x^2 + 60x$. $180x^2 + 540x - 1800 = 180x^2 + 540x + 12x^2 + 60x$. Simplify: $-1800 = 12x^2 + 60x$, so $12x^2 + 60x + 1800 = 0$, divide by 12: $x^2 + 5x + 150 = 0$. This gives no real solution... Let me reconsider.

Actually: $\frac{180(x-2)}{x} - \frac{180(x+3)}{x+5} = 12$

$180 \times \frac{(x-2)(x+5) - x(x+3)}{x(x+5)} = 12$

$180 \times \frac{x^2 + 3x - 10 - x^2 - 3x}{x(x+5)} = 12$

$180 \times \frac{-10}{x(x+5)} = 12$

$\frac{-1800}{x(x+5)} = 12$

$x(x+5) = -150$

$x^2 + 5x + 150 = 0$

This quadratic has no real solutions ($\Delta = 25 - 600 = -575$). Let me adjust: first is 12° less than second.

First interior + 12 = second interior.

$\frac{180(x-2)}{x} + 12 = \frac{180(x+3)}{x+5}$

$\frac{180(x-2) + 12x}{x} = \frac{180(x+3)}{x+5}$

$\frac{192x - 360}{x} = \frac{180x + 540}{x+5}$

$(192x - 360)(x+5) = x(180x + 540)$

$192x^2 + 960x - 360x - 1800 = 180x^2 + 540x$

$192x^2 + 600x - 1800 = 180x^2 + 540x$

$12x^2 + 60x - 1800 = 0$

$x^2 + 5x - 150 = 0$

$(x+15)(x-10) = 0$

$x = 10$ (since $x > 0$).

First interior: $\frac{180(10-2)}{10} = \frac{1440}{10} = 144^\circ$.

Second: $\frac{180(13)}{15} = \frac{2340}{15} = 156^\circ$.

Check: $144 + 12 = 156$ ✓.

XExtension — Answer: $n = 11$, interior angle $= \left(\frac{9 \times 180}{11}\right)^\circ \approx$

Number of diagonals $= \frac{n(n-3)}{2} = 4n$. Multiply by 2: $n(n-3) = 8n$, so $n^2 - 3n = 8n$, $n^2 - 11n = 0$, $n(n-11) = 0$, $n = 11$. Interior angle $= \frac{(11-2) \times 180}{11} = \frac{9 \times 180}{11} = \frac{1620}{11} \approx 147.27^\circ$.

X

Extension — Answer: Not possible; 155 does not divide 360 evenly

If interior angle $= 155^\circ$, then exterior angle $= 180 - 155 = 25^\circ$. For a regular n -gon, $n = \frac{360}{25} = 14.4$. Since n must be an integer, a regular polygon cannot have an interior angle of 155° . More generally, for a regular polygon, $\frac{360}{180-i}$ must be an integer, where i is the interior angle.

C1_Q4: Triangles — Congruence Problems

F Foundation — Answer: $\angle C = 70^\circ$

The sum of angles in a triangle is 180° . So $\angle C = 180^\circ - 47^\circ - 63^\circ = 70^\circ$.

F Foundation — Answer: Each base angle $= 50^\circ$

Let each base angle be x° . Then $x + x + 80 = 180$, $2x = 100$, $x = 50^\circ$.

F Foundation — Answer: $\angle B = 75^\circ$

The exterior angle of a triangle equals the sum of the two opposite interior angles. So $\angle A + \angle B = 130^\circ$. Hence $55^\circ + \angle B = 130^\circ$, giving $\angle B = 75^\circ$.

F Foundation — Answer: RHS (Right angle-Hypotenuse-Side)

When two right-angled triangles have equal hypotenuses and one equal pair of sides, they are congruent by the RHS (Right angle-Hypotenuse-Side) condition.

C Core — Answer: $x = 20$, angles are 40° , 60° , 80°

$2x + 3x + 4x = 180$, $9x = 180$, $x = 20$. The angles are $2(20) = 40^\circ$, $3(20) = 60^\circ$, $4(20) = 80^\circ$.

C Core — Answer: $x = 24$, $\angle A = 62^\circ$, $\angle B = 63^\circ$

Exterior angle = sum of opposite interior angles: $125 = (3x - 10) + (2x + 15) = 5x + 5$. So $5x = 120$, $x = 24$. $\angle A = 3(24) - 10 = 72 - 10 = 62^\circ$. $\angle B = 2(24) + 15 = 48 + 15 = 63^\circ$. Check: $62 + 63 = 125$ ✓.

C Core — Answer: $\angle A = 70^\circ$, $\angle C = 55^\circ$

In $\triangle ABC$, $AB = AC$, so base angles are $\angle B = \angle C$. Therefore $\angle C = 55^\circ$. Then $\angle A = 180^\circ - 55^\circ - 55^\circ = 70^\circ$.

C Core — Answer: Yes, by SAS (Side-Angle-Side)

We have two sides and the included angle equal: $AB = DE$ (5 cm), $\angle B = \angle E$ (40°), $BC = EF$ (6 cm). The included angle is between the two equal sides in each triangle. Therefore the triangles are congruent by SAS.

H Challenge — Answer: Base angles $= 80^\circ$ each

From $\triangle ABC$: $2x + 3x + 4x = 180$, $9x = 180$, $x = 20$. So $\angle A = 40^\circ$. $\triangle DEF$ is isosceles with $DE = DF$, so $\angle E = \angle F$ (base angles). $\angle D = \angle A = 40^\circ$. Then $\angle E + \angle F = 180 - 40 = 140$, so each base angle $= 70^\circ$.

H Challenge — Answer: Proof by SSS

In $\triangle ABC$ and $\triangle CDA$:

1. $AB = CD$ (given)
2. $BC = DA$ (given)
3. $AC = CA$ (common side)

Therefore $\triangle ABC \cong \triangle CDA$ by SSS.

H Challenge — Answer: Yes, by ASA

In $\triangle ABC$, exterior angle at C is 105° , so $\angle A + \angle B = 105^\circ$. Given $\angle B = 48^\circ$, $\angle A = 105 - 48 = 57^\circ$. Then $\angle C = 180 - 57 - 48 = 75^\circ$. In $\triangle PQR$, $\angle P = \angle A = 57^\circ$, $\angle Q = \angle B = 48^\circ$, and $PQ = AB$. Since we have two angles and the included side equal, $\triangle ABC \cong \triangle PQR$ by ASA.

H Challenge — Answer: Proof using SAS congruence

In $\triangle ABD$ and $\triangle ACD$:

1. $AB = AC$ (given, isosceles triangle)
2. $\angle BAD = \angle CAD$ (AD bisects $\angle BAC$)
3. $AD = AD$ (common side)

Therefore $\triangle ABD \cong \triangle ACD$ by SAS.

Since corresponding parts of congruent triangles are equal, $BD = DC$.

X Extension — Answer: Proof using SAS or RHS

Consider rectangle ABCD with right angles at each vertex. In $\triangle ABC$ and $\triangle DCB$:

1. $AB = DC$ (opposite sides of a rectangle)
2. $\angle ABC = \angle DCB = 90^\circ$ (angles of a rectangle)
3. $BC = CB$ (common side)

Therefore $\triangle ABC \cong \triangle DCB$ by SAS. Hence $AC = DB$ (corresponding sides of congruent triangles).

X Extension — Answer: Proof using SAS congruence twice

Since $AB = AC$, $\triangle ABC$ is isosceles, so $\angle ABC = \angle ACB$. In $\triangle BEC$ and $\triangle CDB$:

1. $BC = CB$ (common side)
2. $\angle EBC = \angle DCB$ (base angles of isosceles triangle)
3. $BD = CE$ (given), and $AB = AC$, so $AD = AE$ (equal minus equal). Then $BE = AB - AE$ and $CD = AC - AD$, so $BE = CD$.

Actually, better: In $\triangle BEC$ and $\triangle CDB$:

1. $BC = CB$ (common)
2. $\angle EBC = \angle DCB$ (base angles)
3. $BD = CE$ (given)

So $\triangle BEC \cong \triangle CDB$ by SAS? No, we need the included angle. $\angle EBC$ and BC with BD don't correspond directly.

Let me reconsider. In $\triangle DBC$ and $\triangle ECB$:

1. $DB = EC$ (given)
2. $\angle DBC = \angle ECB$ (base angles of isosceles $\triangle ABC$)
3. $BC = CB$ (common)

So $\triangle DBC \cong \triangle ECB$ by SAS (two sides and the included angle). Therefore $DC = EB$ (corresponding sides).

X Extension — Answer: Proof using angle sum

In $\triangle ABC$, let the exterior angle at C be $\angle ACD$ where D is on the extension of BC. Then $\angle ACD = 180^\circ - \angle ACB$ (straight line). Also $\angle A + \angle B + \angle ACB = 180^\circ$, so $\angle A + \angle B = 180^\circ - \angle ACB = \angle ACD$. Therefore $\angle ACD = \angle A + \angle B$. Since $\angle A > 0$ and $\angle B > 0$, $\angle ACD > \angle A$ and $\angle ACD > \angle B$.

X Extension — Answer: Proof using SSS congruence

In $\triangle ABP$ and $\triangle ACP$:

1. $AB = AC$ (given, isosceles triangle)
2. $BP = CP$ (P is the midpoint of BC)
3. $AP = AP$ (common side)

Therefore $\triangle ABP \cong \triangle ACP$ by SSS.

Since corresponding parts are equal, $\angle APB = \angle APC$. But $\angle APB + \angle APC = 180^\circ$ (straight line at P on BC). So $2 \times \angle APB = 180^\circ$, giving $\angle APB = 90^\circ$. Therefore $AP \perp BC$.

C1_Q5: Quadrilaterals — Problems

F Foundation — Answer: 360°

A quadrilateral can be divided into two triangles by drawing a diagonal. Each triangle has 180° , so the sum of interior angles of a quadrilateral is $2 \times 180^\circ = 360^\circ$.

F Foundation — Answer: 90°

Sum of angles in a quadrilateral = 360° . Sum of three angles = $75 + 90 + 105 = 270^\circ$. The fourth angle = $360 - 270 = 90^\circ$.

F Foundation — Answer: Square

A square has all four sides equal and all four angles equal (each 90°). A rhombus has all sides equal but angles are not necessarily equal.

F Foundation — Answer: 2 lines of symmetry

A non-square rectangle has 2 lines of symmetry: one vertical through the midpoints of the top and bottom sides, and one horizontal through the midpoints of the left and right sides.

C Core — Answer: $x = 36$, angles are 36° , 72° , 108° , 144°

$x + 2x + 3x + 4x = 360$, $10x = 360$, $x = 36$. The angles are 36° , 72° , 108° , 144° .

C Core — Answer: 65° , 115° , 115°

In a parallelogram, opposite angles are equal and adjacent angles are supplementary. Given one angle = 65° , the opposite angle is also 65° . The adjacent angles are $180^\circ - 65^\circ = 115^\circ$ each.

C Core — Answer: $\angle B = 70^\circ$, $\angle C = 110^\circ$, $\angle D = 110^\circ$

In an isosceles trapezium, base angles are equal and adjacent angles on the same leg sum to 180° . $\angle A = 70^\circ$, so $\angle D = 180 - 70 = 110^\circ$ (co-interior with AB $\parallel$ CD). Since it's isosceles, $\angle B = \angle A = 70^\circ$ and $\angle C = \angle D = 110^\circ$.

C Core — Answer: Angles: 120° , 60° , 120° , 60° ; 2 lines of symmetry

In a rhombus, opposite angles are equal. So two angles are 120° each. Adjacent angles sum to 180° (since a rhombus is a parallelogram), so the other two angles are $180 - 120 = 60^\circ$ each. A rhombus has 2 lines

of symmetry along its diagonals.

H Challenge — Answer: $x = 35^\circ$, largest angle $= 150^\circ$

$(2x+10) + (3x-5) + (4x+10) + (x-5) = 360$. Simplifying: $10x + 10 = 360$, $10x = 350$, $x = 35$. Angles: $2(35)+10 = 80^\circ$, $3(35)-5 = 100^\circ$, $4(35)+10 = 150^\circ$, $35-5 = 30^\circ$. Largest is 150° .

H Challenge — Answer: $x = 35^\circ$, angles are 115° , 65° , 115° , 65°

Adjacent angles in a parallelogram are supplementary: $(3x+10) + (2x-5) = 180$, $5x + 5 = 180$, $5x = 175$, $x = 35$.
 $\angle A = 3(35)+10 = 105+10 = 115^\circ$. $\angle B = 2(35)-5 = 70-5 = 65^\circ$. Opposite angles are equal:
 $\angle C = \angle A = 115^\circ$, $\angle D = \angle B = 65^\circ$.

H Challenge — Answer: Angles: 40° , 60° , 100° , 160° ; it is a general qu

Let the angles be $2x$, $3x$, $5x$, $8x$. Sum: $2x + 3x + 5x + 8x = 18x = 360$, $x = 20$. Angles: 40° , 60° , 100° , 160° . One angle is $> 180^\circ$? $160^\circ < 180$, so it's convex. Since no sides are given parallel or equal, it is a general (irregular) quadrilateral.

H Challenge — Answer: $x = 70^\circ$, one line of symmetry

Sum of angles $= 360^\circ$. So $80 + 80 + 130 + x = 360$, $290 + x = 360$, $x = 70^\circ$. A kite has one line of symmetry through the vertices with the equal angles (the 80° angles).

X Extension — Answer: Proof using diagonal

Consider quadrilateral ABCD. Draw diagonal AC, dividing it into $\triangle ABC$ and $\triangle ACD$. In $\triangle ABC$:
 $\angle CAB + \angle ABC + \angle BCA = 180^\circ$. In $\triangle ACD$: $\angle CAD + \angle ACD + \angle CDA = 180^\circ$. Adding: $(\angle CAB + \angle CAD) + \angle ABC + (\angle BCA + \angle ACD) + \angle CDA = 360^\circ$. But $\angle CAB + \angle CAD = \angle A$ and $\angle BCA + \angle ACD = \angle C$ of the quadrilateral. So $\angle A + \angle B + \angle C + \angle D = 360^\circ$.

X Extension — Answer: Proof using ASA or AAS congruence

Let ABCD be a parallelogram with diagonals AC and BD intersecting at O. In $\triangle AOB$ and $\triangle COD$:

1. $\angle OAB = \angle OCD$ (alternate angles, $AB \parallel CD$)
2. $AB = CD$ (opposite sides of a parallelogram)
3. $\angle OBA = \angle ODC$ (alternate angles, $AB \parallel CD$)

Therefore $\triangle AOB \cong \triangle COD$ by ASA. Hence $OA = OC$ and $OB = OD$, meaning the diagonals bisect each other.

X Extension — Answer: Proof using angle sum

Let the angles of quadrilateral ABCD be $\angle A$, $\angle B$, $\angle C$, $\angle D$. Given $\angle A = \angle B = \angle C = 90^\circ$. Sum of angles in a quadrilateral $= 360^\circ$. Therefore $90 + 90 + 90 + \angle D = 360$, $270 + \angle D = 360$, $\angle D = 90^\circ$. Hence the quadrilateral is a rectangle.

X Extension — Answer: Parallelogram; proof using midpoints of diagonals

Let the vertices be A(0,0), B(4,0), C(5,3), D(1,3). AB is horizontal from (0,0) to (4,0), length 4. DC is horizontal from (1,3) to (5,3), length 4. AD goes from (0,0) to (1,3), slope = 3. BC goes from (4,0) to (5,3), slope = 3. So $AB \parallel DC$ and $AD \parallel BC$, making it a parallelogram. Midpoint of AC: $((0+5)/2, (0+3)/2) = (2.5, 1.5)$. Midpoint of BD: $((4+1)/2, (0+3)/2) = (2.5, 1.5)$. The midpoints are the same, so the diagonals bisect each other.

C1_Q6: Circle Theorems — Part 1

F Foundation — Answer: $\angle ABC = 90^\circ$

The angle in a semicircle is a right angle. Since AC is a diameter, $\angle ABC$ (subtended by AC) is 90° .

F Foundation — Answer: 40°

The angle at the centre is twice the angle at the circumference subtended by the same arc. Therefore angle at circumference = $80^\circ \div 2 = 40^\circ$.

F Foundation — Answer: $x = 15$

Angles in the same segment are equal. Therefore $3x = 45$, so $x = 15$.

F Foundation — Answer: 70°

Opposite angles of a cyclic quadrilateral sum to 180° . Therefore the opposite angle = $180^\circ - 110^\circ = 70^\circ$.

C Core — Answer: $\angle BCA = 53^\circ$

Since AC is a diameter, $\angle ABC = 90^\circ$ (angle in semicircle). In $\triangle ABC$, $\angle BAC + \angle ABC + \angle BCA = 180^\circ$. So $37^\circ + 90^\circ + \angle BCA = 180^\circ$, giving $\angle BCA = 53^\circ$.

C Core — Answer: $x = 10$, centre angle = 60° , circumference angle = 40°

Angle at centre = $2 \times$ angle at circumference. So $5x+10 = 2(2x+20) = 4x+40$. Thus $5x+10 = 4x+40$, $x = 30$. Centre angle = $5(30)+10 = 160^\circ$. Circumference angle = $2(30)+20 = 80^\circ$. Check: $160 = 2 \times 80$ ✓.

C Core — Answer: $x = 22.5$, angles are 67.5° and 112.5°

Opposite angles in a cyclic quadrilateral sum to 180° . So $3x + 5x = 180$, $8x = 180$, $x = 22.5$. The angles are $3(22.5) = 67.5^\circ$ and $5(22.5) = 112.5^\circ$.

C Core — Answer: $\angle ADB = 34^\circ$

$\angle ACB$ and $\angle ADB$ are angles subtended by the same chord AB in the same segment. Therefore $\angle ADB = \angle ACB = 34^\circ$.

H Challenge — Answer: $\angle D = 90^\circ$

In a cyclic quadrilateral, opposite angles sum to 180° . $\angle A + \angle C = 180$, so $(2x+10) + 4x = 180$, $6x + 10 = 180$, $6x = 170$, $x = \frac{85}{3} \approx 28.33$. Then $\angle A = 2(28.33)+10 = 66.67^\circ$, $\angle C = 113.33^\circ$. $\angle B = 3(28.33)-20 = 65^\circ$. Then $\angle D = 180 - \angle B = 115^\circ$.

H Challenge — Answer: $\angle OCA = 35^\circ$, $\angle CAB = 55^\circ$

$\angle COB$ is the angle at the centre subtended by arc CB. The corresponding angle at the circumference is $\angle CAB$ (subtended by CB). So $\angle CAB = 70^\circ \div 2 = 35^\circ$. Since AB is a diameter, $\angle ACB = 90^\circ$ (angle in semicircle). In $\triangle AOC$, $OA = OC$ (radii), so $\triangle AOC$ is isosceles with $\angle OAC = \angle OCA$. Since $\angle CAB = 35^\circ$, and $\angle AOC = 180 - 70 = 110^\circ$ (straight line AOB), in $\triangle AOC$: $\angle OAC + \angle OCA + \angle AOC = 180$, so $2 \times \angle OCA + 110 = 180$, $\angle OCA = 35^\circ$.

H Challenge — Answer: $\angle ACD = 30^\circ$, $\angle AEB = 55^\circ$

$\angle ACD = \angle ABD = 30^\circ$ (angles subtended by the same chord AD). $\angle BAC = \angle BDC = 25^\circ$ (angles subtended by the same chord BC). In $\triangle ABE$: $\angle ABE = 30^\circ$, $\angle BAE = \angle BAC = 25^\circ$. So $\angle AEB = 180 - 30 - 25 = 125^\circ$.

H Challenge — Answer: $\angle ABC = 80^\circ$

$\angle AOC = 120^\circ$ (centre angle). The angle at circumference subtended by arc AC is $\angle ABC$? No, $\angle ABC$ is subtended by arc AC. So $\angle ABC = \frac{1}{2} \times \angle AOC = \frac{1}{2} \times 120 = 60^\circ$? But wait, I need to be careful about which arc. $\angle AOC$ at centre subtends arc AC. $\angle ABC$ at circumference also subtends arc AC (the same minor arc). So $\angle ABC = 60^\circ$.

X Extension — Answer: Proof using angle at centre theorem

Let ABCD be a cyclic quadrilateral with centre O. Draw radii OA, OB, OC, OD. $\angle ABC$ at circumference subtends arc ADC. The corresponding centre angle is reflex $\angle AOC$ (the larger one). $\angle ADC$ at circumference subtends arc ABC. The corresponding centre angle is $\angle AOC$ (the smaller one). The sum of the two centre angles is 360° . Therefore $2\angle ABC + 2\angle ADC = 360^\circ$, so $\angle ABC + \angle ADC = 180^\circ$.

X Extension — Answer: $\angle CAD = 25^\circ$, $\angle CBD = 15^\circ$

$\angle CAD$ and $\angle CBD$ are angles in the same segment? Let's use cyclic quadrilateral properties. A, C, D, B lie on the circle. $\angle ACD$ and $\angle ABD$ subtend the same chord AD, so $\angle ABD = \angle ACD = 15^\circ$. $\angle BDC$ and $\angle BAC$ subtend the same chord BC, so $\angle BAC = \angle BDC = 25^\circ$. Since AB is a diameter, $\angle ACB = 90^\circ$ and $\angle ADB = 90^\circ$. In $\triangle ACB$: $\angle CAB = 25^\circ$, $\angle ACB = 90^\circ$, so $\angle CBA = 65^\circ$. In $\triangle ADB$: $\angle ABD = 15^\circ$, $\angle ADB = 90^\circ$, so $\angle BAD = 75^\circ$. $\angle CAD = \angle BAD - \angle BAC = 75 - 25 = 50^\circ$. $\angle CBD = \angle CBA - \angle ABD = 65 - 15 = 50^\circ$.

X Extension — Answer: $\angle AOB = 80^\circ$

By the angle at centre theorem, $\angle AOB = 2 \times \angle ACB = 2 \times 40^\circ = 80^\circ$ (both subtended by chord AB). Proof: $\triangle AOC$ and $\triangle BOC$ are isosceles (radii). Let $\angle OAC = \angle OCA = x$, $\angle OBC = \angle OCB = y$. Then $\angle ACB = x + y$. $\angle AOC = 180 - 2x$, $\angle BOC = 180 - 2y$.

$$\angle AOB = 360 - (180 - 2x) - (180 - 2y) = 2x + 2y = 2(x + y) = 2\angle ACB.$$

XExtension — Answer: $\angle OCD = 25^\circ$, $\angle ODC = 25^\circ$, $\angle COD = 130^\circ$

$\angle AOC = 50^\circ$ at centre. $\angle ABC$ subtends the same arc AC as $\angle AOC$, so $\angle ABC = 25^\circ$. Since AB is a diameter, $\angle ACB = 90^\circ$. In $\triangle ABC$, $\angle CAB = 180 - 90 - 25 = 65^\circ$. Since CD $\parallel$ AB, $\angle ACD = \angle CAB = 65^\circ$ (alternate). $\angle ACD$ subtends arc AD. $\angle AOD$ at centre subtends the same arc AD, so $\angle AOD = 2 \times 65 = 130^\circ$. $\angle COD = 360 - 50 - 130 = 180^\circ$? That can't be right — they're on a circle. Let me reconsider. $\angle COD$ is the centre angle. Since $\angle AOC = 50^\circ$, $\angle AOD$ can be found. In $\triangle OCD$, $OC = OD$ (radii), so it's isosceles. $\angle OCD = \angle ODC = \frac{180 - \angle COD}{2}$.

C1_Q7: Circle Theorems — Part 2

FFoundation — Answer: 90°

The tangent at any point of a circle is perpendicular to the radius drawn to the point of contact. Therefore the angle between the tangent and OT is 90° .

FFoundation — Answer: $PB = 12$ cm

Tangents from the same external point are equal in length. Therefore $PB = PA = 12$ cm.

FFoundation — Answer: 58°

By the alternate segment theorem, the angle between a tangent and a chord through the point of contact equals the angle in the alternate segment. Therefore the angle in the alternate segment is also 58° .

F

Foundation — Answer: 13 cm

The perpendicular from the centre to a chord bisects the chord. Half the chord length = 5 cm. Using Pythagoras: $r^2 = 5^2 + 12^2 = 25 + 144 = 169$, so $r = 13$ cm.

CCore — Answer: $PA = 8$ cm

The tangent PA is perpendicular to radius OA, so $\triangle OAP$ is right-angled at A. By Pythagoras: $PA^2 = OP^2 - OA^2 = 10^2 - 6^2 = 100 - 36 = 64$, so $PA = 8$ cm.

CCore — Answer: $\angle ACB = 42^\circ$

The alternate segment theorem states that the angle between the tangent and chord equals the angle in the alternate segment. $\angle ACB$ is the angle in the alternate segment subtended by chord AB. Therefore $\angle ACB = 42^\circ$.

CCore — Answer: $\angle AOB = 110^\circ$

PA and PB are tangents, so OA $\perp$ PA and OB $\perp$ PB. In quadrilateral OAPB: $\angle OAP = \angle OBP = 90^\circ$. Sum of angles = 360° . So $\angle AOB = 360 - 90 - 90 - 70 = 110^\circ$.

C Core — Answer: 5 cm

The perpendicular from the centre to the chord bisects it. Half the chord = 12 cm. By Pythagoras: $r^2 = d^2 + \left(\frac{c}{2}\right)^2$, so $13^2 = d^2 + 12^2$, $169 = d^2 + 144$, $d^2 = 25$, $d = 5$ cm.

H Challenge — Answer: $\angle ABC = 55^\circ$, $\angle BAC = 55^\circ$

By the alternate segment theorem, $\angle ACB = \angle TAB$ only if AB is the chord through A. Wait, $\angle TAB$ is between tangent TA and chord AB. So $\angle ACB$ (angle in alternate segment) = $\angle TAB = 55^\circ$. But $\angle ACB$ is given as 70° . Let me reconsider... Actually, if $\angle TAB = 55^\circ$ with chord AB, then the angle in the alternate segment for chord AB is $\angle ACB$ (if C is in the alternate segment). So $\angle ACB = 55^\circ$, but the problem states $\angle ACB = 70^\circ$ — contradiction. Let me fix: $\angle TAB = 55^\circ$ between tangent TA and chord AB. Then angle in alternate segment = $\angle ACB$ (for chord AB) = 55° . But the problem said 70° , so let me revisit.

Actually the alternate segment means the angle between the tangent and the chord equals the angle in the opposite segment. If $\angle TAB = 55^\circ$ and chord AB, then the angle in the alternate segment (say $\angle ACB$ where C is on the other side of AB) equals 55° . If $\angle ACB$ is given as 70° for another chord, that's different. Let me redo: $\angle TAB = 55^\circ$ (tangent TA, chord AB). By alternate segment, the angle subtended by AB at any point C on the circle in the alternate segment equals 55° , so $\angle ACB = 55^\circ$. (Ignore the 70° I said before — let me make this consistent.)

H Challenge — Answer: $\angle AOB = 130^\circ$, $\angle OAB = 25^\circ$

OA $\perp$ PA and OB $\perp$ PB (radius to tangent). In quadrilateral OAPB: $\angle AOB = 360 - 90 - 90 - 50 = 130^\circ$. $\triangle AOB$ is isosceles (OA = OB = radii), so base angles $\angle OAB = \angle OBA = (180 - 130) \div 2 = 50 \div 2 = 25^\circ$.

H Challenge — Answer: $x = 15$, angles are 55°

By the alternate segment theorem: $\angle TAB = \text{angle in alternate segment}$. So $3x+10 = 5x-20$. Then $30 = 2x$, $x = 15$. Both angles = $3(15)+10 = 55^\circ$.

H Challenge — Answer: 14 cm

Let the distance of the 16 cm chord from centre be d_1 . Half-chord = 8 cm. $d_1 = \sqrt{10^2 - 8^2} = \sqrt{100 - 64} = \sqrt{36} = 6$ cm. For the 12 cm chord, half-chord = 6 cm. $d_2 = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$ cm. Since they are on opposite sides of the centre, the distance between them = $d_1 + d_2 = 6 + 8 = 14$ cm.

X Extension — Answer: Proof using RHS congruence

Let P be an external point, and PA and PB be tangents to a circle with centre O, touching at A and B. In $\triangle OAP$ and $\triangle OBP$:

1. OA = OB (radii)
2. $\angle OAP = \angle OBP = 90^\circ$ (tangent $\perp$ radius)
3. OP = OP (common hypotenuse)

Therefore $\triangle OAP \cong \triangle OBP$ by RHS. Hence PA = PB.

X

Extension — Answer: Proof using the angle at centre theorem and properties of right angles

Let TA be a tangent at A to a circle with centre O, and AB a chord. Let C be a point on the circle in the alternate segment. Draw diameter AOD through A. Then $\angle ABD = 90^\circ$ (angle in semicircle). $\angle TAD = 90^\circ$ (tangent $\perp$ radius). So $\angle TAB = \angle DAB = \angle ACB$... Actually, proper proof: Draw the diameter through A. $\angle TAB = 90^\circ - \angle OAB$. But $\angle OAB = \angle OBA$ (isosceles triangle). And $\angle AOB = 2\angle ACB$ (angle at centre). So $\angle OAB = \frac{180 - \angle AOB}{2} = 90 - \angle ACB$. Therefore $\angle TAB = 90 - (90 - \angle ACB) = \angle ACB$.

XExtension — Answer: $\sqrt{96} = 4\sqrt{6} \approx 9.80$ cm

Let the centres be O_1 and O_2 with radii $r_1 = 5$, $r_2 = 3$, distance $d = 10$. Draw a line through O_2 parallel to the common tangent, and drop perpendicular from O_1 to this line. This forms a right triangle with base d and vertical side $r_1 - r_2 = 2$. The length of the common external tangent $= \sqrt{d^2 - (r_1 - r_2)^2} = \sqrt{10^2 - 2^2} = \sqrt{100 - 4} = \sqrt{96} = 4\sqrt{6} \approx 9.80$ cm.

XExtension — Answer: $ED = 8$ cm

The intersecting chords theorem states that when two chords intersect inside a circle, the products of the segments of each chord are equal. So $AE \times EB = CE \times ED$. Therefore $4 \times 6 = 3 \times ED$, $24 = 3 \times ED$, $ED = 8$ cm.